Water Wheel — Teacher Guide

Flagship Lab 2 · Grade 4 Using Natural Resources for Energy · Teacher

Name: _____ **Date:** _____

Thursday: hands-on testing. Friday: analysis, redesign, reflection.

BIG QUESTION

How does falling water transfer energy to lift a load?

CONNECTION TO REAL SYSTEMS

Real hydroelectric systems use spinning turbines connected to generators. In this lab, the spinning wheel (turbine) lifts a load instead of turning a generator — the energy pathway is the same.

FALLING WATER → **SPINNING WHEEL** → **LIFTED LOAD**

VARIABLES

TYPE	DETAIL
Independent (change)	pour height (ONE variable)
Response (measure)	paperclips lifted (count)
Constants (keep same)	wheel design · axle · blade setup · water volume · pour opening · thread/load system

MATERIALS (PER GROUP OF 4)

paper plate or plastic lid wheel · 8–10 blade cups/popsicle-stick blades · straw or dowel axle · ~50 cm thread · paperclips (load) · small cup with a pour opening · 500 mL water source · basin/tray to catch water · ruler (height) · towel

SETUP & SAFETY

- Build the wheel first (blades on a plate/lid, axle through the center).
- Hang thread from the axle; attach paperclips as the load.
- Pour water from a cup with a fixed opening onto the blades over a basin.
- Change ONLY pour height (e.g., 15 cm, 30 cm, 45 cm). Same water volume (500 mL) and same pour opening each trial.
- Safety: keep water over the basin; wipe spills immediately; no horseplay near water.

TIMING (30-MIN CLASS)

SEGMENT	MINUTES
Build wheel + review fair test	10
Test 3 pour heights (rotate jobs)	15
Record + quick cleanup	5

EXPECTED OUTCOME

Higher pour height → falling water has more energy of motion → wheel spins faster/longer → lifts more paperclips (or lifts the fixed load in less time). Do NOT pre-print these results on the student sheet; students infer them.

WHAT TO LISTEN FOR

- Students keep the wheel/water/pour the same and change only height.
- Students use 'energy of motion' and 'transfer' (not 'mechanical energy').
- Students connect falling water → spinning wheel → lifted load to dams.

ABSENT-STUDENT RECOVERY

Send the student sheet + a partner's data table. Have the absent student complete the claim, pathway, and redesign using the provided data.

ASSESSMENT

Formative: fair-test check, data table, claim, redesign, check-your-thinking response.